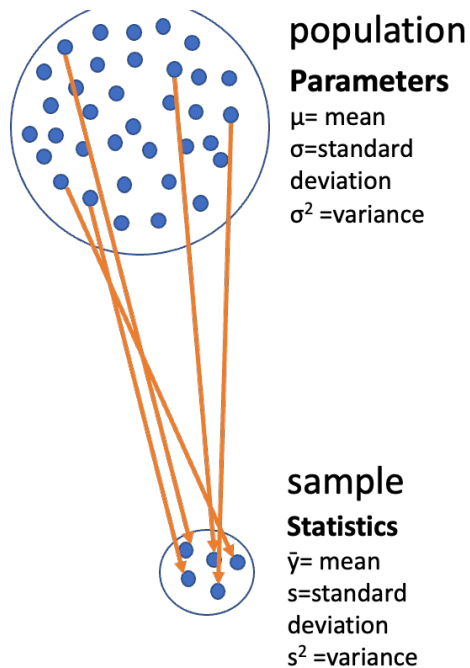


Lecture 05: Probability and Statistical Inference

Bill Perry

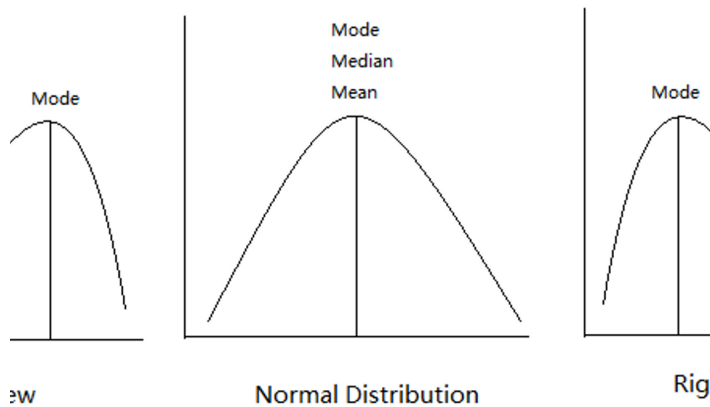
Lecture 4: Review

- Introduction to histograms or frequency distributions
- Probability Distribution Functions (PDF)
- Descriptive Statistics
 - Center - mean, median, mode
 - Spread - range, variance, standard deviation
- Tests of means using T-Tests
 - one sample
 - two sample



Lecture 4: Review - Statistical Concepts

- Introduction to histograms or frequency distributions
- Probability Distribution Functions (PDF)
- Descriptive Statistics
 - Center - mean, median, mode
 - Spread - range, variance, standard deviation
- Tests of means using T-Tests
 - one sample
 - two sample



Lecture 4: Review - Summary Statistics

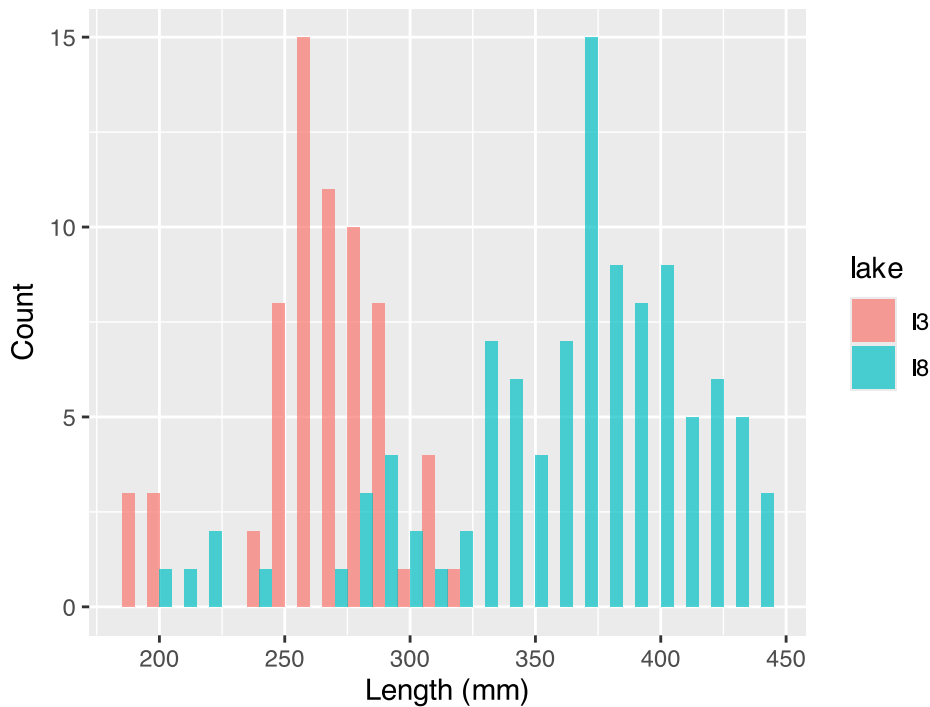
- Introduction to histograms or frequency distributions
- Probability Distribution Functions (PDF)
- Descriptive Statistics
 - Center - mean, median, mode
 - Spread - range, variance, standard deviation
- Tests of means using T-Tests
 - one sample - is the sample mean different from a hypothesized mean?
 - two sample - are the sample means from two samples the same or different?

```
# A tibble: 2 × 5
  lake mean_length sd_length se_length count
<chr>   <dbl>    <dbl>    <dbl> <int>
1 I3    266.    28.3     3.48    66
2 I8    363.    52.3     5.18   102
```

Lecture 5: Probability and Statistical Inference

The goals for today

- Statistical inference fundamentals
- Hypothesis testing principles
- T Distributions
- One sample T Tests
- Two sample T Test



Arctic grayling length data from lakes I3 and I8

Lecture 5: Student's t-distribution Table

Here is a t-table

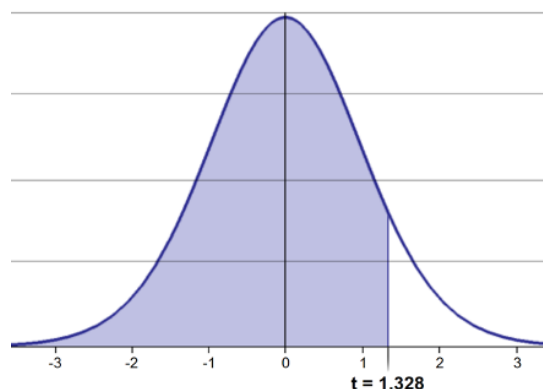
- Values of t that correspond to probabilities
- Probabilities listed along top
- Sample dfs are listed in the left-most column
- Probabilities are given for one-tailed and two-tailed “questions”

one-tail	0.50	0.25	0.20	0.15	0.10	0.05	0.025	0.01	0.005	0.001	0.0005
two-tails	1.00	0.50	0.40	0.30	0.20	0.10	0.05	0.02	0.01	0.002	0.001
df											
1	0.000	1.000	1.376	1.963	3.078	6.314	12.71	31.82	63.66	318.31	636.62
2	0.000	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	0.000	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	0.000	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.000	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.000	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.000	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	0.000	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	0.000	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	0.000	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	0.000	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	0.000	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	0.000	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	0.000	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	0.000	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	0.000	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	0.000	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	0.000	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	0.000	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	0.000	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21	0.000	0.686	0.859	1.063	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	0.000	0.686	0.858	1.061	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	0.000	0.685	0.858	1.060	1.319	1.714	2.069	2.500	2.807	3.485	3.768
24	0.000	0.685	0.857	1.059	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	0.000	0.684	0.856	1.058	1.316	1.708	2.060	2.485	2.787	3.450	3.725
26	0.000	0.684	0.856	1.058	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27	0.000	0.684	0.855	1.057	1.314	1.703	2.052	2.473	2.771	3.421	3.690
28	0.000	0.683	0.855	1.056	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	0.000	0.683	0.854	1.055	1.311	1.699	2.045	2.462	2.756	3.396	3.659
30	0.000	0.683	0.854	1.055	1.310	1.697	2.042	2.457	2.750	3.385	3.646
40	0.000	0.681	0.851	1.050	1.303	1.684	2.021	2.423	2.704	3.307	3.551
60	0.000	0.679	0.848	1.045	1.296	1.671	2.000	2.390	2.660	3.232	3.460
80	0.000	0.678	0.846	1.043	1.292	1.664	1.990	2.374	2.639	3.195	3.416
100	0.000	0.677	0.845	1.042	1.290	1.660	1.984	2.364	2.626	3.174	3.390
1000	0.000	0.675	0.842	1.037	1.282	1.646	1.962	2.330	2.581	3.098	3.300
Z	0.000	0.674	0.842	1.036	1.282	1.645	1.960	2.326	2.576	3.090	3.291
	0%	50%	60%	70%	80%	90%	95%	98%	99%	99.8%	99.9%

Lecture 5: One-tailed Questions

One-tailed questions: area of distribution left or (right) of a certain value

- $n=20$ ($df=19$) - 90% of the observations found left
- $t = 1.328$ (10% are outside)

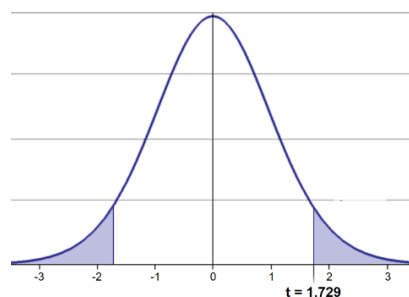


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100	0.000	0.677	0.845	1.042	1.290	1.660	1.984	2.364	2.626	3.174	3.390

Lecture 5: Two-tailed Questions

Two-tailed questions refer to area between certain values

- $n = 20$ ($df=19$), 90% of the observations are between
- $t=-1.729$ and $t=1.729$ (10% are outside)



5	0.000	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.000	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.000	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408
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13	0.000	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221
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17	0.000	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646	3.965
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19	0.000	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579	3.883
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21	0.000	0.686	0.859	1.063	1.323	1.721	2.080	2.518	2.831	3.527	3.819
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28	0.000	0.683	0.855	1.056	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	0.000	0.683	0.854	1.055	1.311	1.699	2.045	2.462	2.756	3.396	3.659
30	0.000	0.683	0.854	1.055	1.310	1.697	2.042	2.457	2.750	3.385	3.646
40	0.000	0.681	0.851	1.050	1.303	1.684	2.021	2.423	2.704	3.307	3.551

Lecture 5: Calculating CI Example

Let's calculate CIs again:

Use two-sided test

- $CI = \bar{y} \pm t \cdot \frac{s}{\sqrt{n}}$
- 95% CI Sample A: $= 272.8 \pm 2.262 * (37.81/(10^{0.5})) = 27.05$
- The 95% CI is between 245.8 and 299.8
- “The 95% CI for the population mean from sample A is 272.8 ± 27.05 ”

1	0.000	1.000	1.376	1.963	3.078	6.314	12.71	31.82	63.66	318.31	636.62
2	0.000	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	0.000	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.215	12.924
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27	0.000	0.684	0.855	1.057	1.314	1.703	2.052	2.473	2.771	3.421	3.690
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80	0.000	0.678	0.846	1.043	1.292	1.664	1.990	2.374	2.639	3.195	3.416
100	0.000	0.677	0.845	1.042	1.290	1.660	1.984	2.364	2.626	3.174	3.390
1000	0.000	0.675	0.842	1.037	1.282	1.646	1.962	2.330	2.581	3.098	3.300

Lecture 5: Applications of t-distribution

So:

- Can assess confidence that population mean is within a certain range
- Can use t distribution to ask questions like:
 - “What is probability of getting sample with mean = $\bar{y}$ from population with mean = μ ?” (1 sample t-test)
 - “What is the probability that two samples came from same population?” (2 sample t-test)

Lecture 5: Single Sample T-Test

We want to test if the mean fish length in I3 differs from 240mm.

Activity: Define hypotheses and identify assumptions

$H_0: \mu = 240$ (The mean fish length in I3 is 240mm)

$H_1: \mu \neq 240$ (The mean fish length in I3 is not 240mm)

Assumptions for t-test:

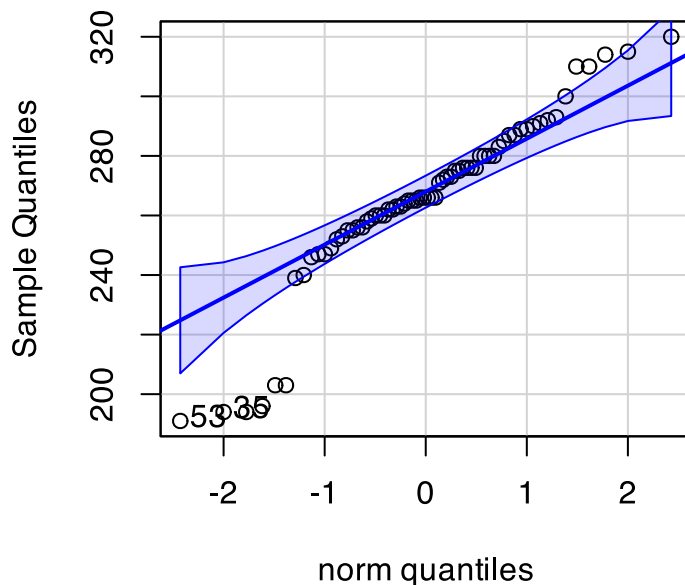
1. Data is normally distributed
2. Observations are independent
3. No significant outliers

Assumptions in R - qqplots from car

```
# Filter for just windward side needles

# YOUR TASK: Test normality of windward pine needle lengths
# QQ Plot
qqPlot(i3_df$length_mm,
       main = "QQ Plot for length of Grayling",
       ylab = "Sample Quantiles")
```

QQ Plot for length of Grayling



```
[1] 53 35
```

Statistical Test of Normality

Shapiro-Wilk test

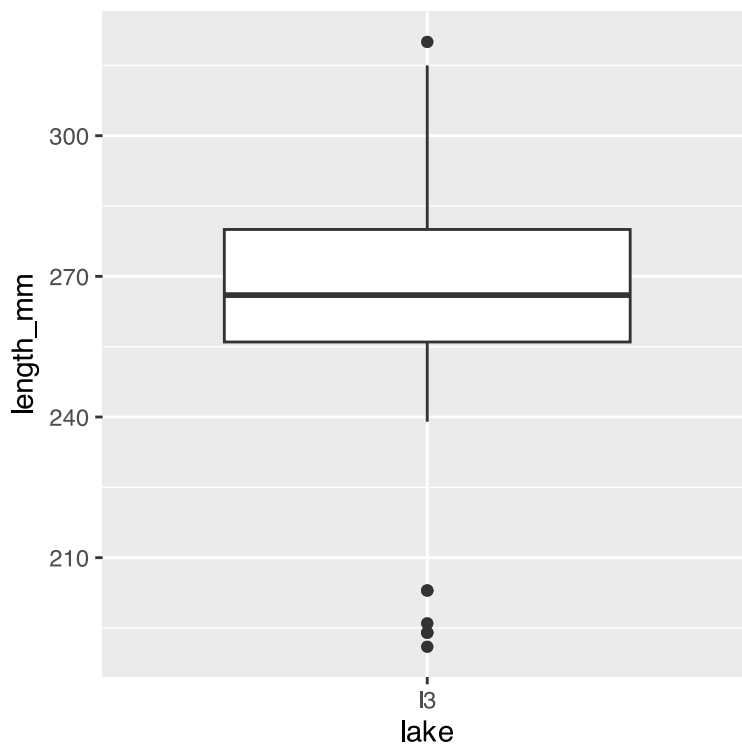
```
# Shapiro-Wilk test
shapiro.test(i3_df$length_mm)
```

Shapiro-Wilk normality test

```
data: i3_df$length_mm
W = 0.91051, p-value = 0.0001623
```

Checking for Outliers

```
# Check for outliers using boxplot
# YOUR CODE HERE
i3_df %>% ggplot(aes(lake, length_mm))+geom_boxplot()
```



Practice Exercise 1: One-Sample t-Test

💡 Practice Exercise 1: One-Sample t-Test

Let's perform a one-sample t-test to determine if the mean fish length in I3 Lake differs from 240 mm:

```
# what is the mean
i3_mean <- mean(i3_df$length_mm, na.rm=TRUE)
cat("Mean:", round(i3_mean, 1), "mm\n")
```

Mean: 265.6 mm

```
# Perform a one-sample t-test
t_test_result <- t.test(i3_df$length_mm, mu = 240)

# View the test results
t_test_result
```

One Sample t-test

```
data: i3_df$length_mm
t = 7.3497, df = 65, p-value = 4.17e-10
alternative hypothesis: true mean is not equal to 240
95 percent confidence interval:
 258.6481 272.5640
sample estimates:
mean of x
 265.6061
```

Interpret this test result by answering these questions:

1. What was the null hypothesis?
2. What was the alternative hypothesis?
3. What does the p-value tell us?
4. Should we reject or fail to reject the null hypothesis at $\alpha = 0.05$?
5. What is the practical interpretation of this result for fish biologists?

Lecture 5: Hypothesis Testing Framework

Hypothesis testing is a systematic way to evaluate research questions using data.

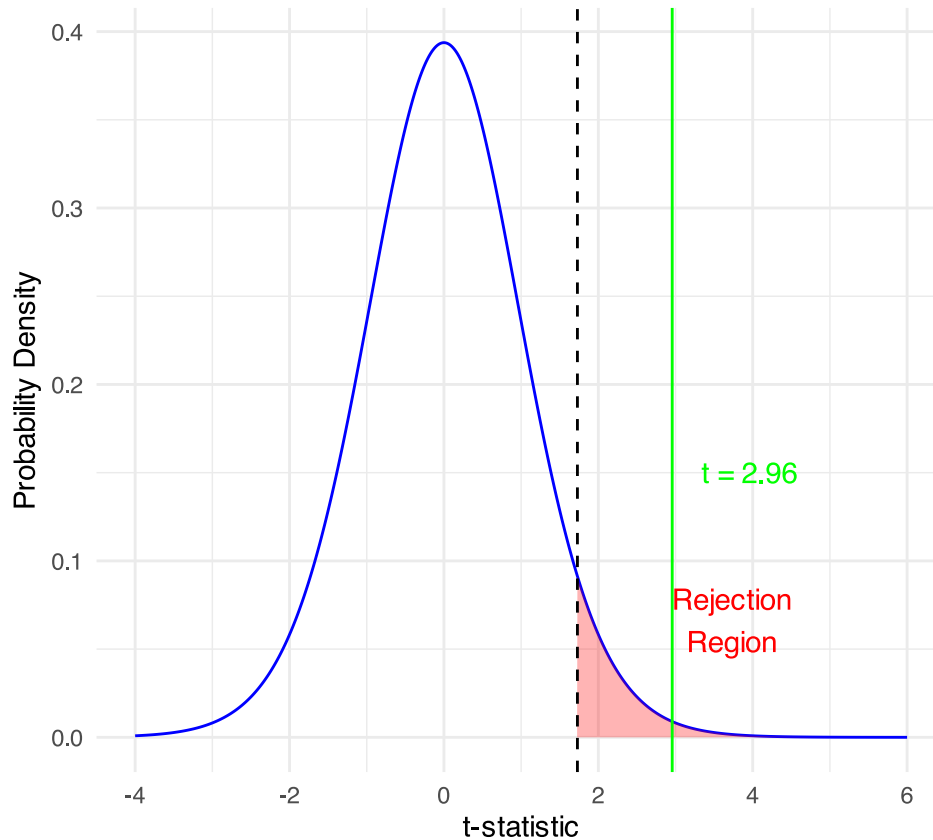
Key components:

1. **Null hypothesis (H_0):** Typically assumes “no effect” or “no difference”
2. **Alternative hypothesis (H_a):** The claim we’re trying to support
3. **Statistical test:** Method for evaluating evidence against H_0
4. **P-value:** Probability of observing our results (or more extreme) if H_0 is true
5. **Significance level (α):** Threshold for rejecting H_0 , typically 0.05

Decision rule: Reject H_0 if p-value less than α or shorthand $p < 0.05$

One-Sample t-Test

$H_0: \mu = 240$ vs $H_1: \mu \neq 240$ ($\alpha = 0.05$, $df = 19$)



Lecture 5: Hypothesis Testing

Hypothesis testing is a systematic way to evaluate research questions using data.

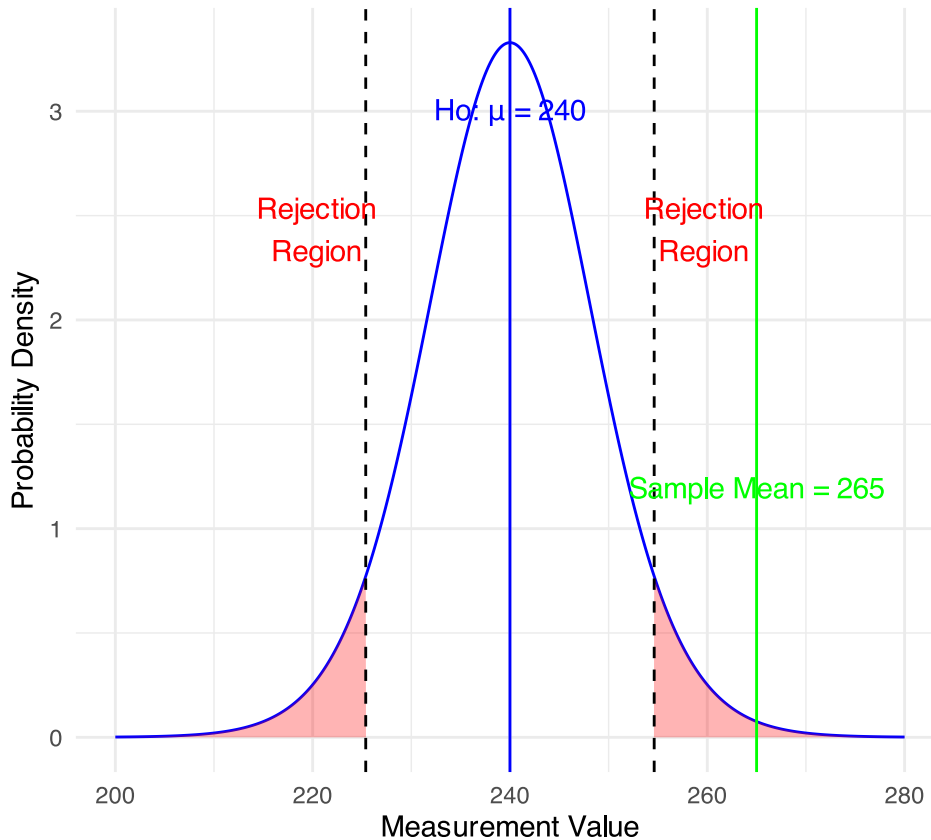
Key components:

1. **Null hypothesis (H_0):** Typically assumes “no effect” or “no difference”
2. **Alternative hypothesis (H_a):** The claim we’re trying to support
3. **Statistical test:** Method for evaluating evidence against H_0
4. **P-value:** Probability of observing our results (or more extreme) if H_0 is true
5. **Significance level (α):** Threshold for rejecting H_0 , typically 0.05

Decision rule: Reject H_0 if $p\text{-value} < \alpha$

One-Sample t-Test in Original Scale

Testing $H_0: \mu = 240$ ($\alpha = 0.05$, $df = 19$)



Lecture 5: Interpreting One-Sample T-Test Results

Activity: Interpret the t-test results

- What does the p-value tell us?
- Should we reject or fail to reject the null hypothesis?

How to report this result in a scientific paper:

“A one-sample t-test at $\alpha=0.05$ showed that the mean fish length (... mm, SD = ...) [was/was not] significantly different from the expected 240 mm, $t(\dots) = \dots$, $p = \dots$ ”

Lecture 5: Two Sample T-Tests Introduction

For example

- what is probability that population X is the same as population Y?

How would you assess this question using what we learned?

This is what we will do with the fish length again...



Lecture 5: Comparing Two Samples

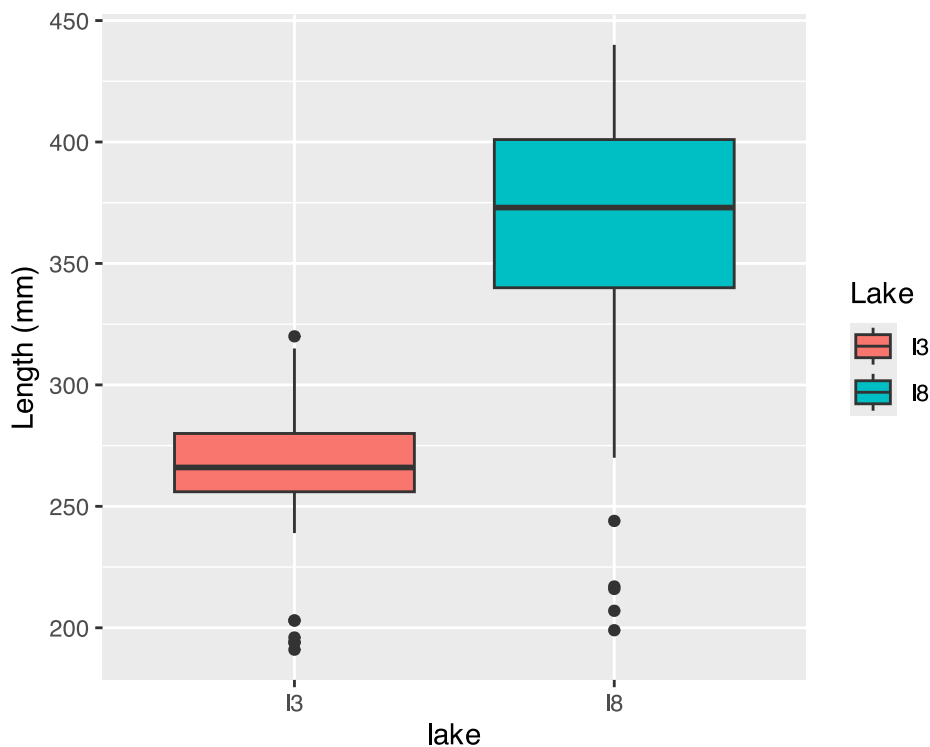
For example

- what is probability that population X is the same as population Y?

How would you assess this question using what we learned?

```
# Now create a boxplot to visualize the difference in fish lengths between these lakes:
```

```
# Create a boxplot comparing the two lakes
i3_i8_plot <- grayling_df %>%
  ggplot(aes(x = lake, y = length_mm, fill = lake)) +
  geom_boxplot() +
  labs(
    x = "lake",
    y = "Length (mm)",
    fill = "Lake")
i3_i8_plot
```



```
# Based on the t-test results and the boxplot
#
# what can you conclude about the fish populations in these two lakes?
```

Practice Exercise 2: Formulating Hypotheses

Practice Exercise 2: Formulating Hypotheses

For the following research questions about fish lengths write the null and alternative hypotheses:

1. Are fish lengths in lake I3 and I8 different?

What are the hypotheses?

Ho =

Ha =

Lecture 5: Two-Sample T-Test Framework

Now, let's compare fish lengths from the two lakes

Question: **Is there a significant difference in fish length between the lakes?**

This requires a two-sample t-test.

Two-sample t-test compares means from two independent groups.

$$t = \frac{\bar{x}_1 - \bar{x}_2}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

where:

- $\bar{x}_1$ and $\bar{x}_2$: These represent the sample means of the two groups you're comparing
- s_p^2 : This is the pooled variance, calculated as: $s_p^2 = [(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2] / (n_1 + n_2 - 2)$, where s_1^2 and s_2^2 are the sample variances of the two groups.
- **n_1 and n_2** : These are the sample sizes of the two groups.
- $\sqrt{(1/n_1 + 1/n_2)}$: This represents the pooled standard error.

$$t = \frac{SIGNAL}{NOISE}$$

Practice Exercise 3: Summary Statistics

💡 Practice Exercise 3: Calculate summary statistics grouped by lake

Before conducting the test, we need to understand the data for each group.

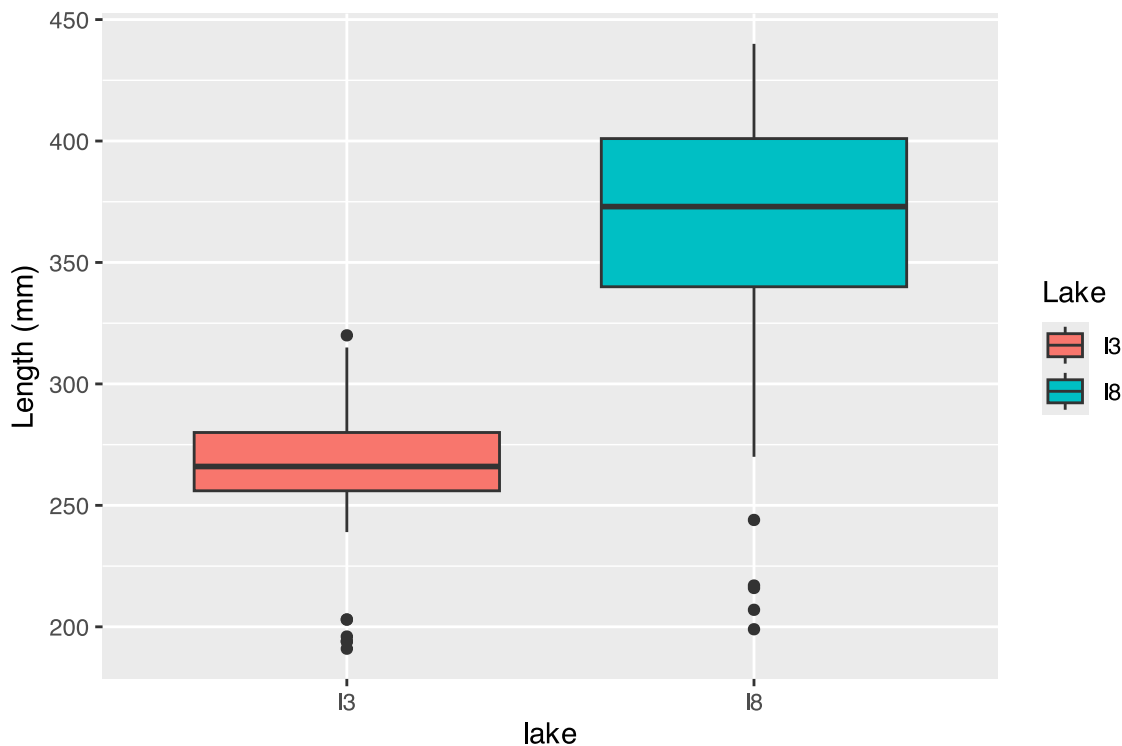
1. You need this and the graph to see what is going on

```
group_summary <- grayling_df %>%  
  group_by(lake) %>%  
  summarize(  
    mean_length = mean(length_mm),  
    sd_length = sd(length_mm),  
    n = n(),  
    se_length = sd_length / sqrt(n)  
  )  
  
print(group_summary)
```

```
# A tibble: 2 × 5  
  lake mean_length sd_length    n se_length  
  <chr>      <dbl>    <dbl> <int>    <dbl>  
1 I3         266.      28.3    66     3.48  
2 I8         363.      52.3   102     5.18
```

Visualizing Group Differences

```
# Create a boxplot comparing the two sides  
i3_i8_plot
```



Practice Exercise 4: Effect Size

💡 Practice Exercise 4: Effect size

We could also look at the difference in means... some cool code here

```
# Assuming your dataframe is called df
group_summary %>%
  summarize(difference = mean_length[lake == "I8"] - mean_length[lake == "I3"])
```

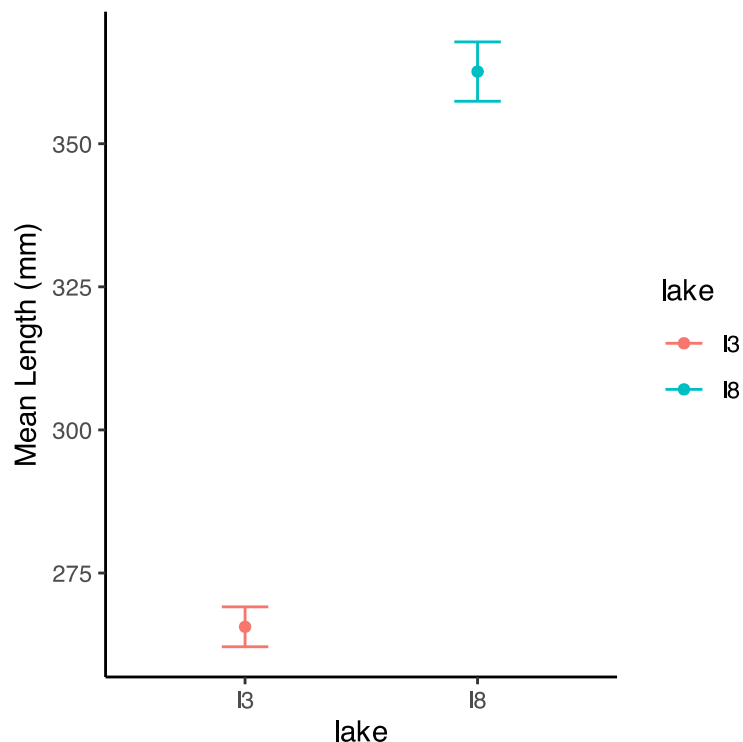
```
# A tibble: 1 × 1
  difference
    <dbl>
1      97.0
```

Practice Exercise 5: ggplot Summary Statistics

💡 Practice Exercise 5: Using GGLOT to get summary stats

GGplot also has code to make the mean and standard error plots we are interested in along with a lot of others

```
# Assuming your dataframe is called df
fish_mean_se_plot <- ggplot(grayling_df, aes(x = lake, y = length_mm, color = lake)) +
  stat_summary(fun = mean, geom = "point") +
  stat_summary(fun.data = mean_se, geom = "errorbar", width = 0.2) +
  labs(
    x = "lake",
    y = "Mean Length (mm)" ) +
  theme_classic()
fish_mean_se_plot
```



Lecture 5: Testing Assumptions for Two-Sample T-Test

For a two-sample t-test, we need to check:

1. Normality within each group
2. Equal variances between groups (for standard t-test)
3. Independent observations

If assumptions are violated:

- Welch's t-test (unequal variances)
- Non-parametric alternatives (Mann-Whitney U test)

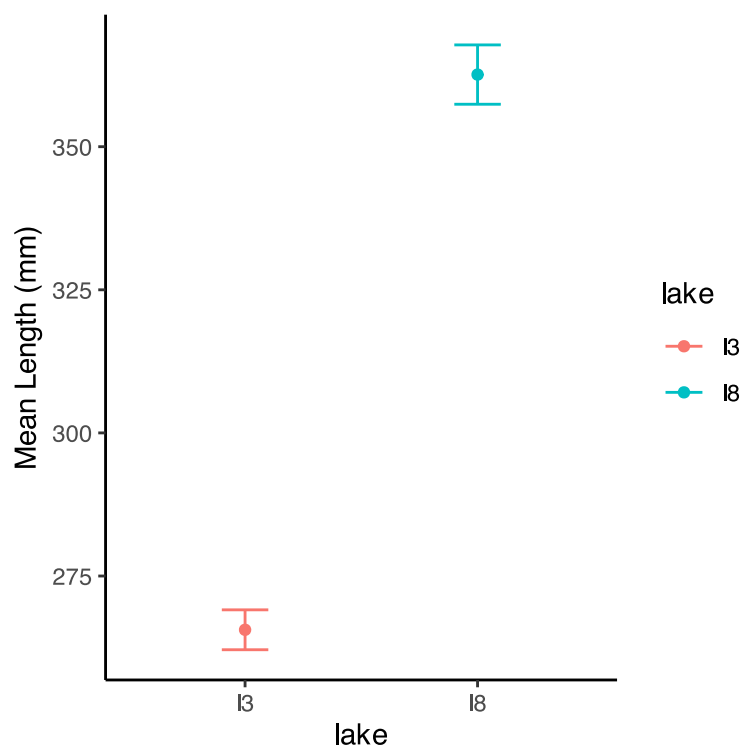
Practice Exercise 6: Creating Group Data

💡 Practice Exercise 6: Test normality of windward pine needle lengths

qqplots

Note you need to test each groups separately...

```
# Assuming your dataframe is called df
fish_mean_se_plot
```



Practice Exercise 7: Separate Group Data

💡 Practice Exercise 7: Test normality of windward pine needle lengths

qqplots

Note you need to test each groups separately...

```
# how do you make separate dataframes to do this on?  
# Separate data by groups  
i3_data <- grayling_df %>% filter(lake == "I3")  
i8_data <- grayling_df %>% filter(lake == "I8")  
head(i8_data)
```

```
# A tibble: 6 × 5  
  site lake species length_mm mass_g  
<dbl> <chr> <chr>      <dbl> <dbl>  
1  118 I8  arctic grayling    345    375  
2  118 I8  arctic grayling    376    485  
3  118 I8  arctic grayling    417    600  
4  118 I8  arctic grayling    369    445  
5  118 I8  arctic grayling    419    625  
6  118 I8  arctic grayling    365    430
```

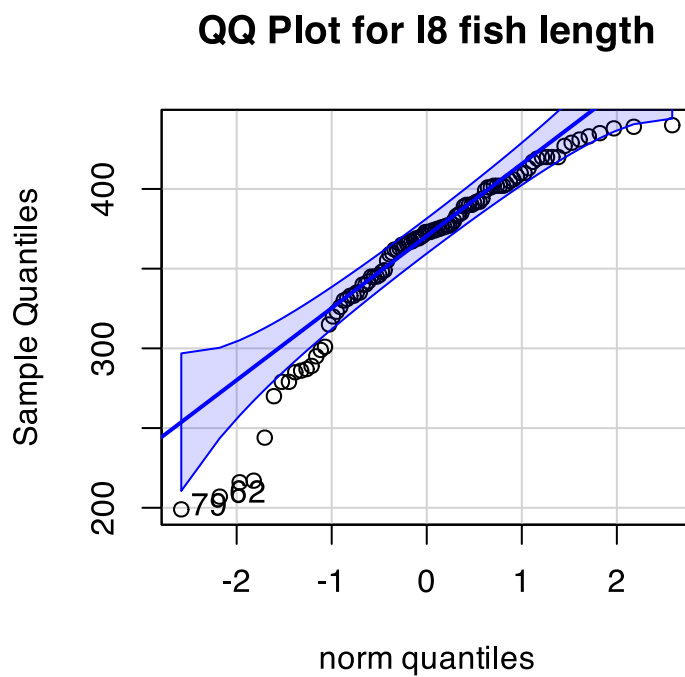
Practice Exercise 8: QQ Plot for Windward Data

💡 Practice Exercise 8: Test normality of windward pine needle lengths

qqplots

Note you need to test each groups separately...

```
# QQ Plot for windward group
qqPlot(i8_data$length_mm,
      main = "QQ Plot for I8 fish length",
      ylab = "Sample Quantiles")
```



```
[1] 79 62
```

Practice Exercise 9: Shapiro-Wilk Test

💡 Practice Exercise 9: Test normality of I8 lengths

Shapiro-Wilk test

Note you need to test each groups separately...

```
# Shapiro-Wilk test for windward group  
shapiro.test(i8_data$length_mm)
```

Shapiro-Wilk normality test

```
data: i8_data$length_mm  
W = 0.91962, p-value = 1.119e-05
```

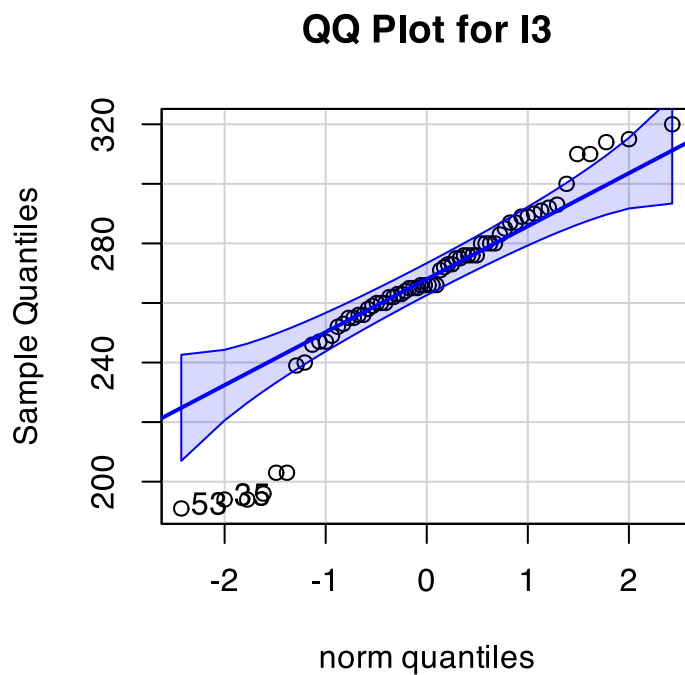
Practice Exercise 10: QQ Plot for I3

💡 Practice Exercise 10: Test normality of I3 lengths

qqplots

Note you need to test each groups separately...

```
# You can also test the i3  
# QQ Plot for leeward group  
qqPlot(i3_data$length_mm,  
       main = "QQ Plot for I3",  
       ylab = "Sample Quantiles")
```



```
[1] 53 35
```

Practice Exercise 11: Shapiro-Wilk for i3

💡 Practice Exercise 11: Test normality of i3 lengths

Shapiro-Wilk test

Note you need to test each groups separately...

```
# Shapiro-Wilk test for leeward group
shapiro.test(i3_data$length_mm)
```

Shapiro-Wilk normality test

```
data:  i3_data$length_mm
W = 0.91051, p-value = 0.0001623
```

Practice Exercise 12: Combined Normality Test

💡 Practice Exercise 12: Test Normality at one time

There are always a lot of ways to do this in R

```
# there are always two ways
# Test for normality using Shapiro-Wilk test for each wind group
# All in one pipeline using tidyverse approach
normality_results <- grayling_df %>%
  group_by(lake) %>%
  summarize(
    shapiro_stat = shapiro.test(length_mm)$statistic,
    shapiro_p_value = shapiro.test(length_mm)$p.value,
    normal_distribution = if_else(shapiro_p_value > 0.05, "Normal", "Non-normal")
  )

# Print the results
print(normality_results)
```

```
# A tibble: 2 × 4
  lake shapiro_stat shapiro_p_value normal_distribution
<chr>    <dbl>         <dbl> <chr>
1 I3      0.911         0.000162 Non-normal
2 I8      0.920         0.0000112 Non-normal
```

Practice Exercise 13: Test Equal Variances

💡 Practice Exercise 13: Test equal variances

Levenes test can be done on the original dataframe

```
# Method 1: Using car package's leveneTest
# This is often preferred as it's more robust to departures from normality
levene_result <- leveneTest(length_mm ~ lake, data = grayling_df)
print("Levene's Test for Homogeneity of Variance:")
```

```
[1] "Levene's Test for Homogeneity of Variance:"
```

```
print(levene_result)
```

```
Levene's Test for Homogeneity of Variance (center = median)
      Df F value    Pr(>F)
group  1  13.705 0.0002907 ***
      166
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Lecture 5: Conducting the Two-Sample T-Test

Now we can compare the mean fish lengths between i3 and i8.

Ho: $\mu_1 = \mu_2$ (The fish lengths do not differ)

Ha: $\mu_1 \neq \mu_2$ (The mean fish lengths differ - direction is not specified)

Deciding between:

- Standard t-test (equal variances)
- Welch's t-test (unequal variances)

Note the Levenes Test should be NOT SIGNIFICANT - What is the null hypothesis

```
Levene's Test for Homogeneity of Variance (center = median)
      Df F value    Pr(>F)
group  1  13.705 0.0002907 ***
      166
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Lecture 5: Running the Two-Sample T-Test

Now we can do a two sample TTEST

Calculate t-statistic manually (optional)

YOUR CODE HERE:

$$t = (\text{mean1} - \text{mean2}) / \sqrt{((s1^2/n1) + (s2^2/n2))}$$

```
grayling_summary
```

```
# A tibble: 2 × 5
  lake mean_length sd_length se_length count
<chr>    <dbl>    <dbl>    <dbl> <int>
1 I3      266.      28.3      3.48    66
2 I8      363.      52.3      5.18   102
```

💡 Tip

```
# YOUR TASK: Conduct a two-sample t-test
# Use var.equal=TRUE for standard t-test or var.equal=FALSE for Welch's t-test

# Standard t-test (if variances are equal)
t_test_result <- t.test(length_mm ~ lake, data = grayling_df, var.equal = TRUE)
print("Standard two-sample t-test:")
```

```
[1] "Standard two-sample t-test:"
```

```
print(t_test_result)
```

Two Sample t-test

```
data: length_mm by lake
t = -13.797, df = 166, p-value < 2.2e-16
alternative hypothesis: true difference in means between group I3 and group I8 is not equal to 0
95 percent confidence interval:
 -110.87187 -83.11208
sample estimates:
mean in group I3 mean in group I8
      265.6061      362.5980
```

```
# Welch's t-test (if variances are unequal)
# YOUR CODE HERE
```

Lecture 5: Difference between a Two-Sample T and Welch's T Test

Standard t-test (Student's t-test)

- Assumes **equal variances** between the two groups being compared
- Uses a **pooled variance estimate** that combines data from both group
- Has **higher statistical power** when the equal variance assumption is met
- **Degrees of freedom** = $n_1 + n_2 - 2$

Welch's t-test

- **Does not assume equal variances** between groups (also called the "unequal variances t-test")
- Uses **separate variance estimates** for each group
- More **robust** when group variances are different
- Uses a more complex **degrees of freedom calculation** (Welch-Satterthwaite equation)
- **Degrees of freedom** are typically non-integer and usually smaller than the standard t-test

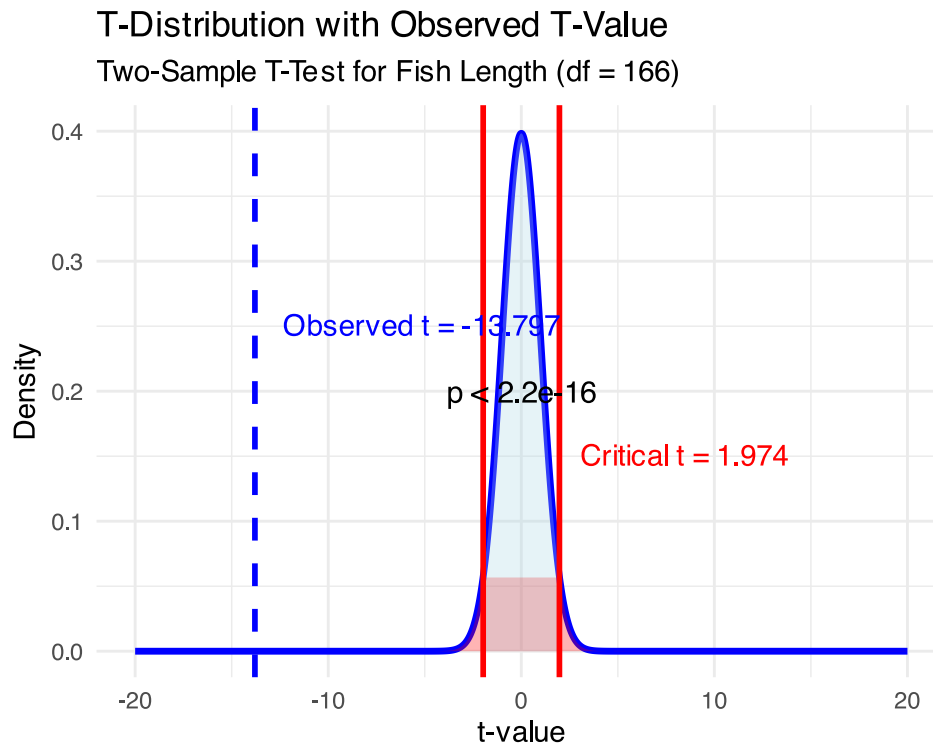
Lecture 5: Interpreting Two-Sample T-Test Results

Interpret the results of the two-sample t-test

What can we conclude about the needle lengths on windward vs. leeward sides?

How to report this result in a scientific paper:

“A two-tailed, two-sample t-test at $\alpha=0.05$ showed [a significant/no significant] difference in needle length between windward ($M = \dots$, $SD = \dots$) and leeward ($M = \dots$, $SD = \dots$) sides of pine trees, $t(\dots) = \dots$, $p = \dots$ ”



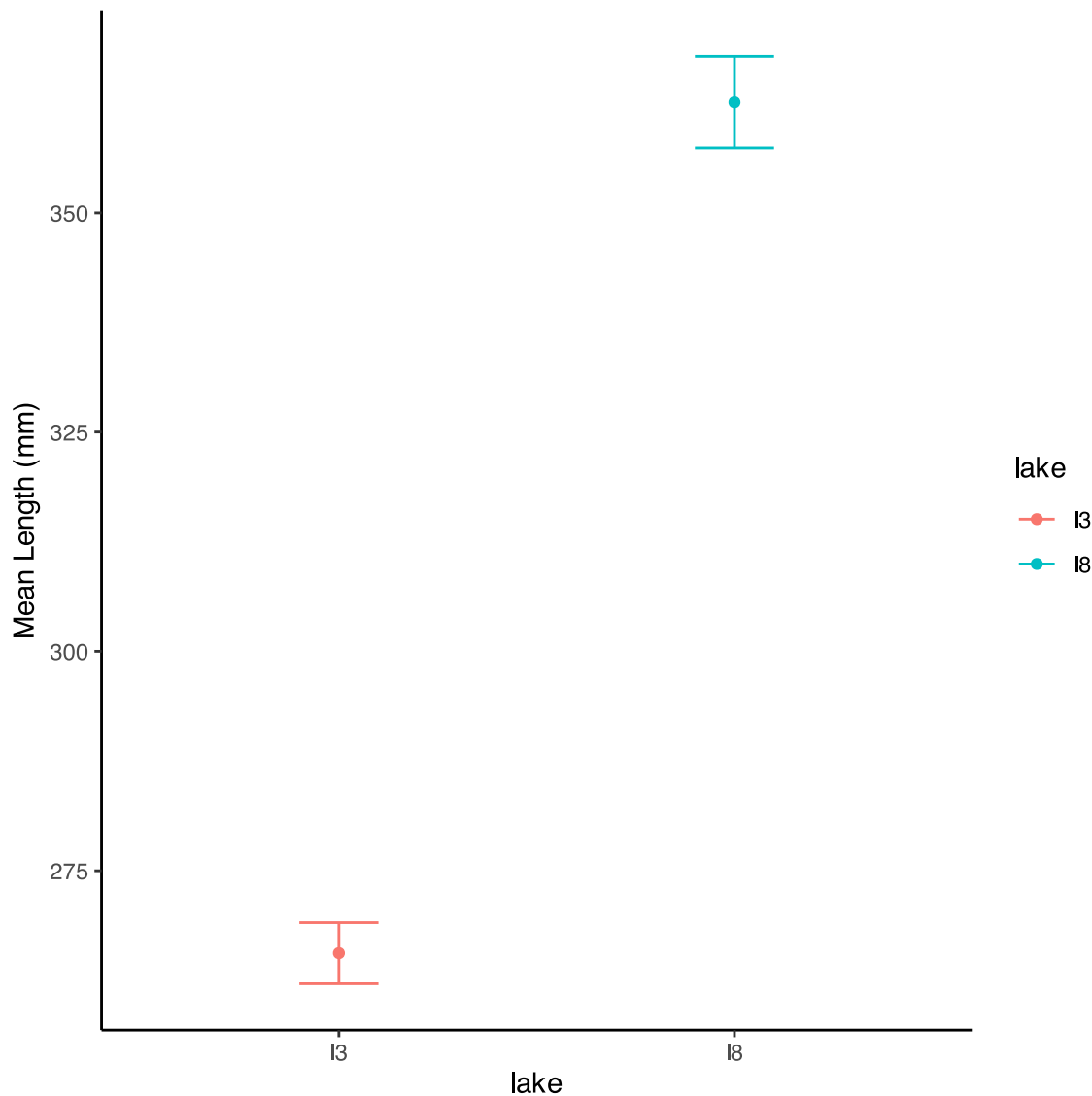
Lecture 5: Visualizing the Results

Interpret the results of the two-sample t-test

What can we conclude about the needle lengths on windward vs. leeward sides?

How to report this result in a scientific paper:

“A two-tailed, two-sample t-test at $\alpha=0.05$ showed [a significant/no significant] difference in needle length between windward ($M = \dots$, $SD = \dots$) and leeward ($M = \dots$, $SD = \dots$) sides of pine trees, $t(\dots) = \dots$, $p = \dots$ ”



Lecture 5: Assumptions of Parametric Tests

Common assumptions for t-tests:

1. Normality: Data comes from normally distributed populations
2. Equal variances (for two-sample tests)
3. Independence: Observations are independent
4. No outliers: Extreme values can influence results

What can we do if our data violates these assumptions?

Alternatives when assumptions are violated:

- Data transformation (log, square root, etc.)
- Non-parametric tests
- Robust statistical methods

Lecture 5: Summary and Conclusions

In this activity, we've:

1. Formulated hypotheses about pine needle length
2. Tested assumptions for parametric tests

3. Conducted one-sample and two-sample t-tests
4. Visualized data using appropriate methods
5. Learned how to interpret and report t-test results

Key takeaways:

- Always check assumptions before conducting tests
- Visualize your data to understand patterns
- Report results comprehensively
- Consider alternatives when assumptions are violated - non parametric tests...